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Data Science- Assignment 4: Model Comparison and Calibration Lab

Dataset Source: Wisconsin Breast Cancer Diagnostic (WDBC)

URL: *https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic*

GitHub: *https://github.com/amsalu2212/DS-Ass4/blob/main/Ass4.ipynb*

Result summary

In this assignment I compared four classical machine learning algorithms based on the available secondary data under one frozen evaluation protocol. The experiment uses the Wisconsin Breast Cancer Diagnostic dataset through scikit-learn's packaged copy, writes a reproducible CSV named amsalu_dataset.csv, and reloads that CSV with pandas. The resulting DataFrame has 569 observations, 30 numeric predictors, and one binary target. There are 357 benign cases and 212 malignant cases with no missing values and no duplicate rows.

Four classical machine-learning models are compared: Logistic Regression, Random Forest, HistGradientBoosting, and a prior-only Dummy classifier. A stratified 80/20 split and shuffled five-fold stratified cross-validation are shared across models. Logistic Regression has the highest mean cross-validation Precision Recall (PR)-AUC (0.993) and lowest mean cross-validation Brier score (0.0201), and it is selected using training cross-validation only. On the holdout set, however, Random Forest has the highest PR-AUC (0.9952) and ROC-AUC (0.9970) among the four models. This holdout result is reported descriptively and does not change the training-CV model selection.

For Logistic Regression, the cost-constrained training-OOF threshold is 0.999612 under a false-negative cost of 5, false-positive cost of 1, and a maximum training alert capacity of 20%. On the holdout set this threshold yields 19 true positives, zero false positives, 23 false negatives, and 72 true negatives (16.7% alert rate; cost 115). The error-slice analysis and most-confident-error counterexample show the consequences of this very restrictive operating threshold. This results are benchmark findings, not clinical validation.

1. Objective and dataset

The objective is to compare a strong traditional baseline, two ensemble classifiers, and a simple reference model under a frozen evaluation protocol. The experiment evaluates discrimination, probability quality, paired model differences, threshold-dependent decisions, and error patterns. The experiment uses classical machine-learning algorithms for this assignment and for practicing exploratory data analysis.

2. Data inspection and exploratory data analysis

Before modelling, the experiment prints the DataFrame type and dimensions, observation and predictor counts, all column names, first and last five rows, data types and non-null counts, per-column missing-value counts, duplicate-row count, descriptive statistics, and target-class counts.

- Class balance: bar chart and pie chart for benign and malignant proportions.
- Feature histograms: mean radius, mean texture, mean perimeter, mean area, worst radius, worst texture, mean smoothness, and mean concavity.
- Class-conditional boxplots: mean radius, mean texture, mean perimeter, and mean area by diagnosis.
- Feature scatterplots: mean radius versus mean texture, and mean perimeter versus mean area, coloured by diagnosis.

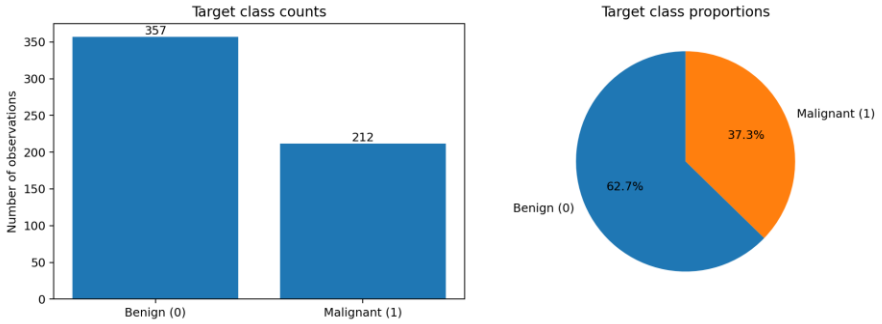


Figure 1. Target class counts and proportions.

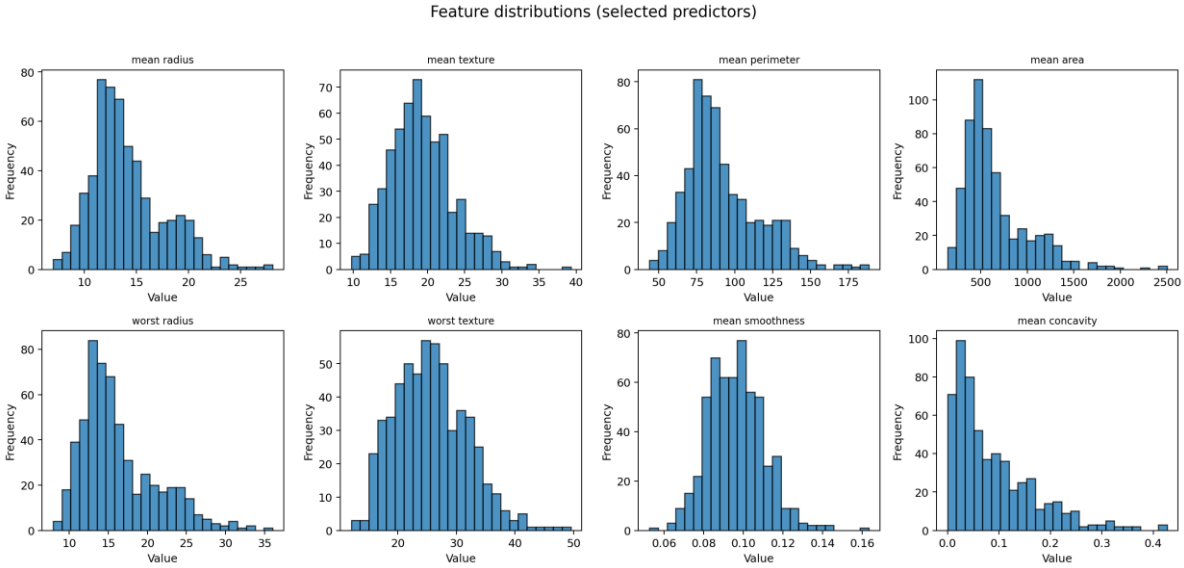


Figure 2. Histograms of selected numeric predictors.

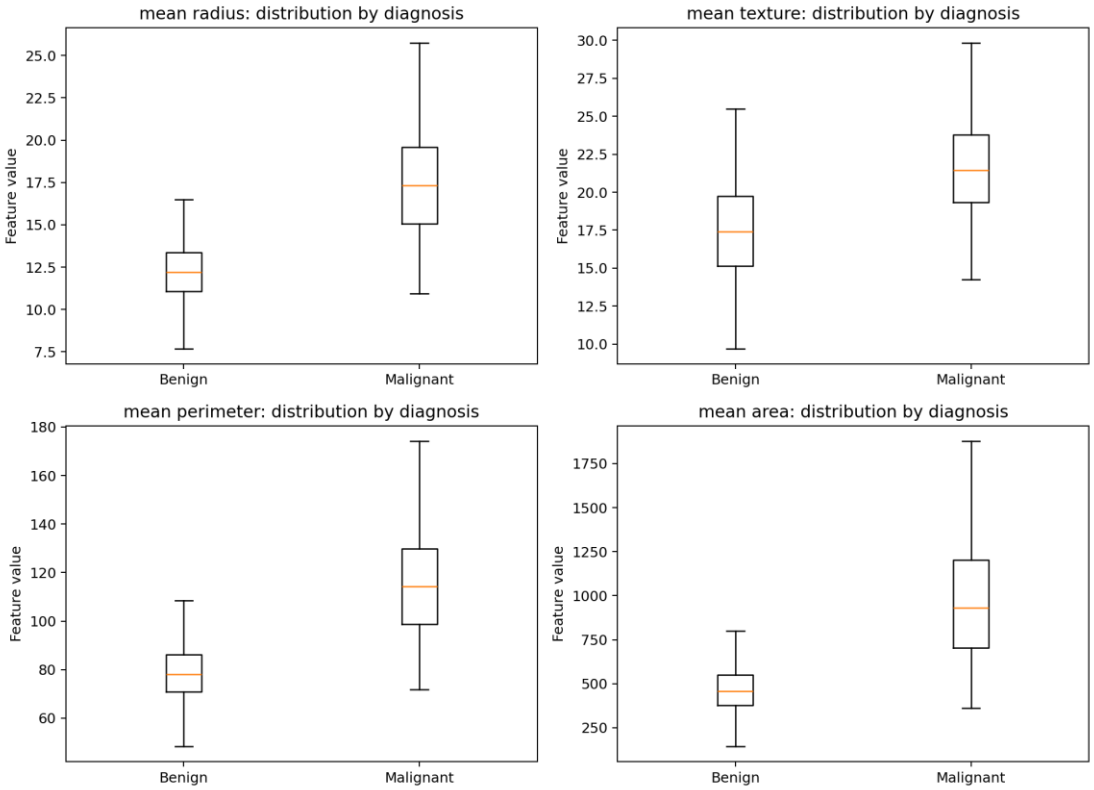


Figure 3. Selected predictor distributions by diagnosis.

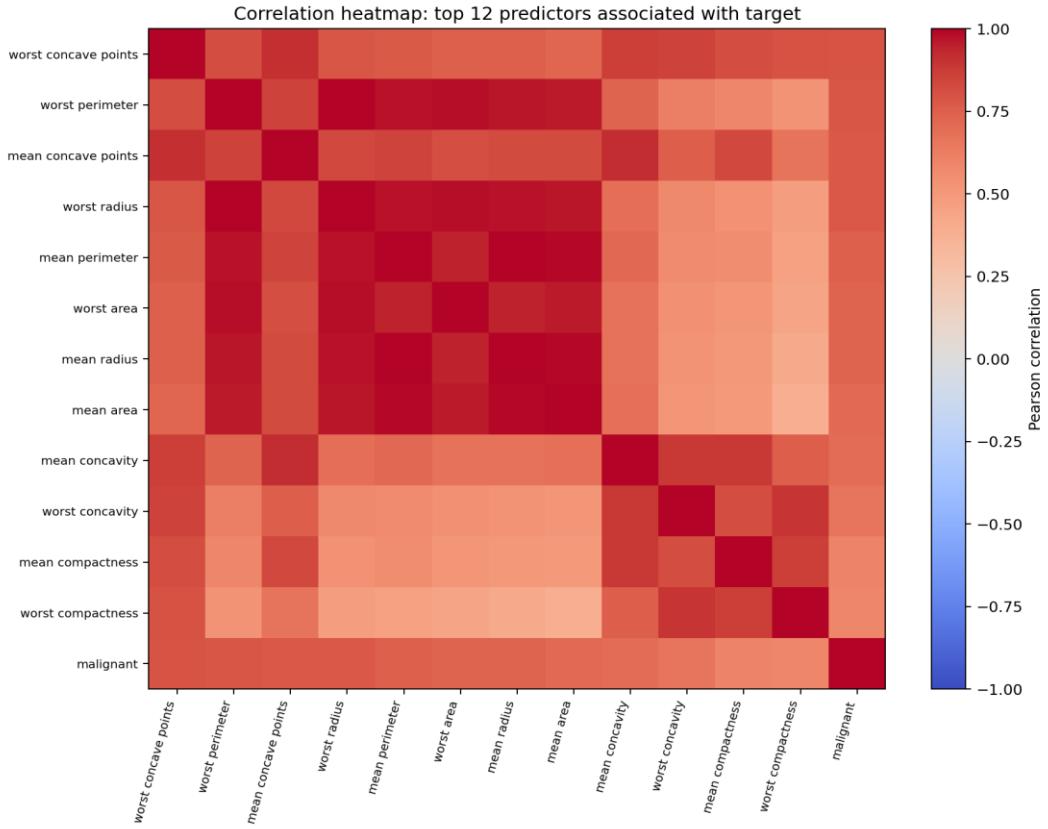


Figure 4. Correlation heatmap of the 12 most target-associated predictors.

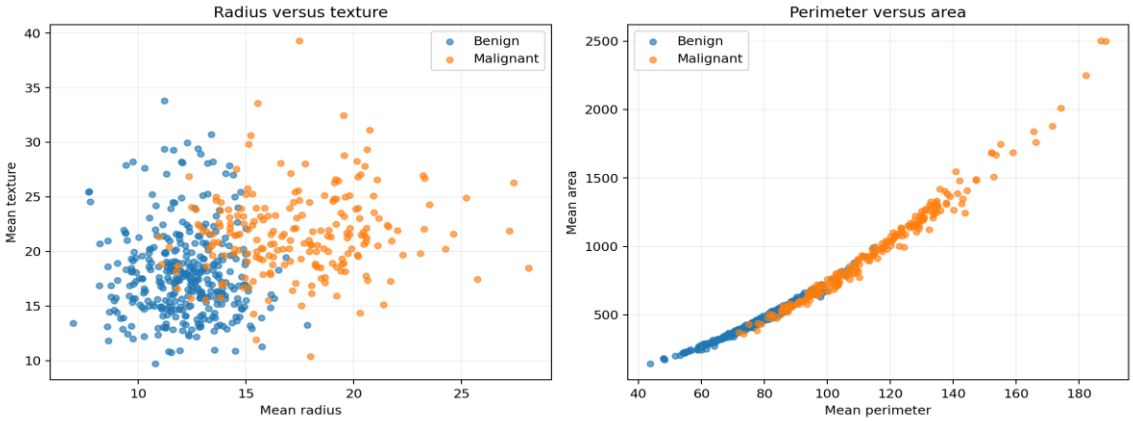


Figure 5. Selected feature relationships, coloured by diagnosis.

These EDA plots are descriptive. They do not change the frozen split or provide independent evidence of performance on a new population.

3. Models and configurations

Model	Purpose	Configuration recorded in experiment
Logistic Regression	Classical, interpretable baseline.	Pipeline: StandardScaler followed by LogisticRegression(C=1.0, max_iter=5000, class_weight='balanced', random_state=42).
Random Forest	Bagging ensemble for nonlinear relationships.	300 trees; max_depth=None; min_samples_leaf=2; class_weight='balanced'; random_state=42; n_jobs=-1.
HistGradientBoosting	Boosting ensemble for nonlinear patterns.	learning_rate=0.08; max_iter=150; max_leaf_nodes=15; l2_regularization=1.0; random_state=42.
Dummy (stratified)	Prior-only reference baseline.	DummyClassifier(strategy='prior').

No hyperparameter search is performed. Each model is evaluated with one fixed configuration and the same data partitions and cross-validation plan.

4. Frozen evaluation protocol

Protocol component	Frozen setting in Experiment
Random seed	42.
Train/holdout partition	Stratified train_test_split; test_size=0.20; random_state=42.
Cross-validation	StratifiedKFold(n_splits=5, shuffle=True, random_state=42).
Bootstrap repetitions	1,000.
Primary metric	PR-AUC / average precision.
Additional metrics	ROC-AUC, Brier score, ECE (10 fixed-width bins), log loss, accuracy, precision, recall, F1, and confusion-matrix counts at threshold 0.50.
Uncertainty summaries	Five-fold t-based confidence intervals for CV summary metrics; 1,000-replicate row-bootstrap intervals for holdout metrics and paired OOF differences.
Threshold costs	FN cost=5.0; FP cost=1.0.
Review capacity	Maximum alert rate=20% on the training OOF predictions.
Selection rule	Select model using highest mean training-CV PR-AUC; choose each model's threshold using training OOF probabilities only.

The holdout partition is reserved for final descriptive evaluation. It is not used to select the model or choose the operational threshold. The code selects Logistic Regression from the cross-validation summary before evaluating the holdout outcomes.

5. Paired predictions and statistics

For each candidate model, the experiment generates out-of-fold probabilities for the same training observations using the same five-fold plan. It computes paired bootstrap differences against Logistic Regression on those same OOF rows, using 1,000 bootstrap replicates. It also saves one holdout probability per model for each identical holdout row.

Model	CV ROC-AUC mean	95% fold t interval	CV PR-AUC mean	95% fold t interval	CV Brier mean
Logistic Regression	0.9954	0.9888–1.0000	0.9944	0.9880–1.0000	0.0201
HistGradientBoosting	0.9931	0.9882–0.9979	0.9907	0.9856–0.9958	0.0273
Random Forest	0.9892	0.9816–0.9967	0.9872	0.9799–0.9945	0.0330
Dummy (stratified)	0.5000	0.5000–0.5000	0.3736	0.3736–0.3736	0.2340

6. Holdout model performance

The holdout set contains 114 observations: 72 benign and 42 malignant. The following metrics are calculated on predicted probabilities; threshold-specific classification metrics use the common threshold 0.50.

Random Forest has the best holdout ROC-AUC and PR-AUC point estimates, but Logistic Regression remains the selected model because the frozen selection rule uses training-CV PR-AUC only. The holdout results must not be used retrospectively to change the selected model.

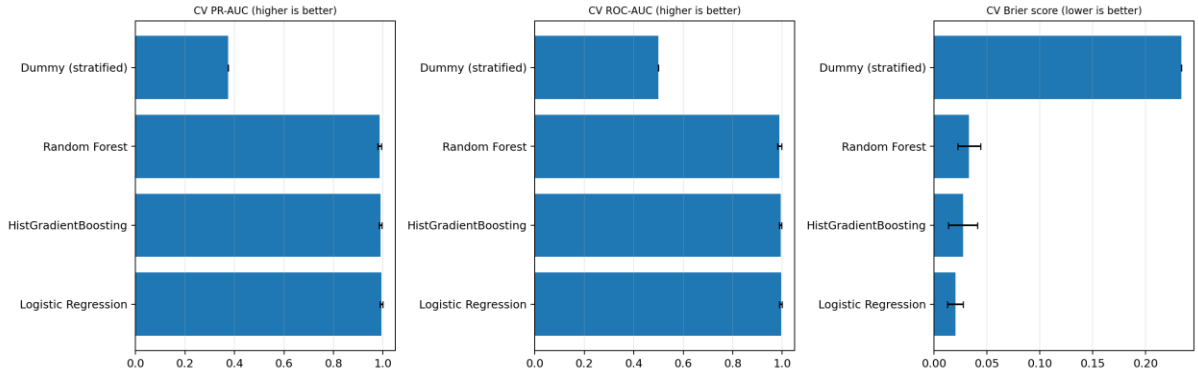


Figure 6. Cross-validation means with fold-based 95% t intervals.

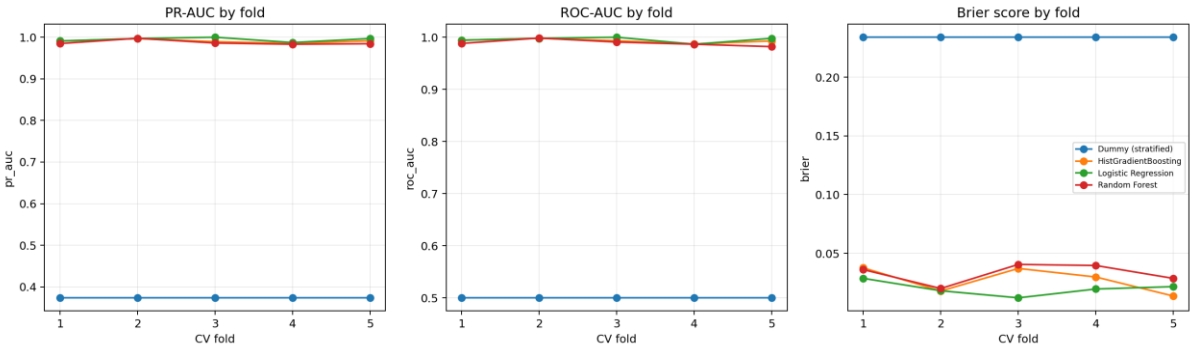


Figure 7. Fold-by-fold metric stability.

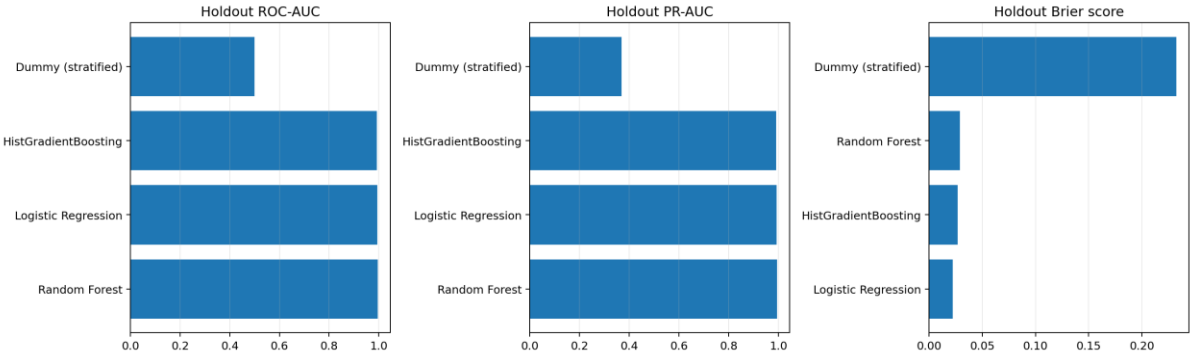


Figure 8. Holdout ROC-AUC, PR-AUC, and Brier comparisons.

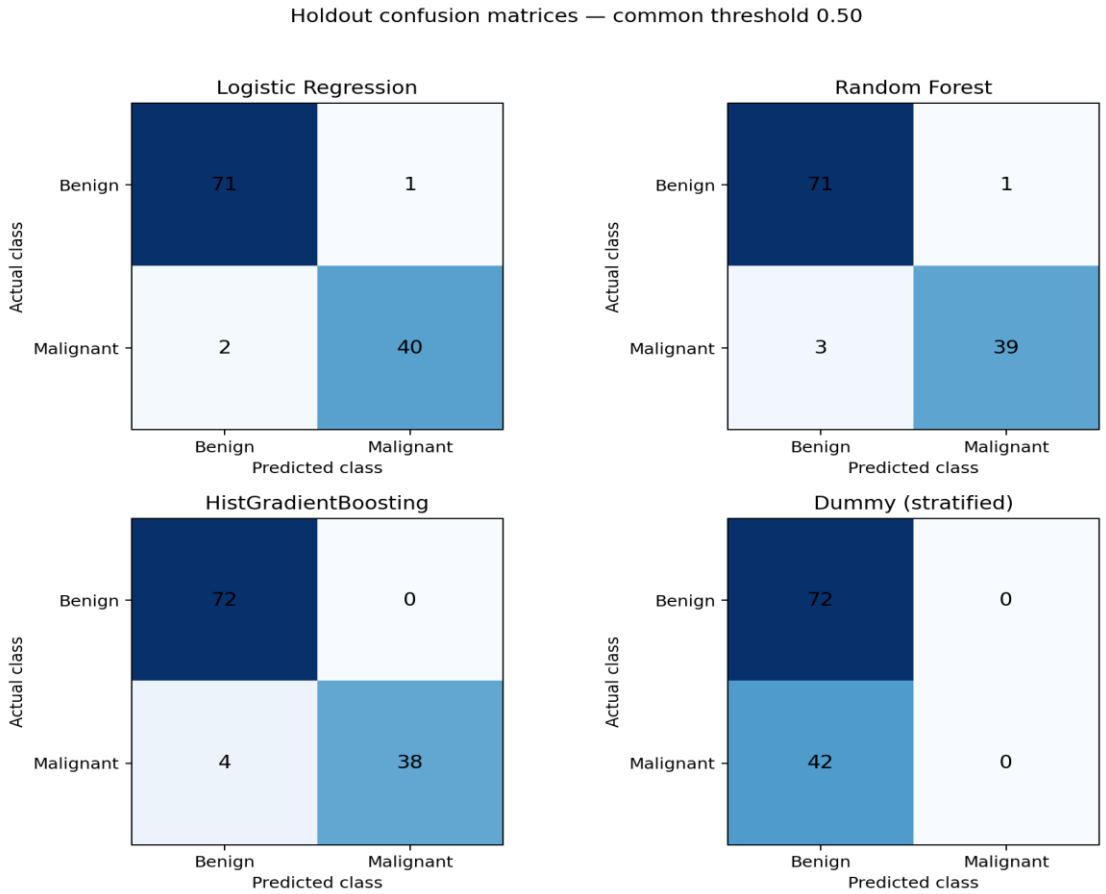


Figure 9. Holdout confusion matrices at the common 0.50 threshold.

7. Calibration and cost curves

Calibration is evaluated using Brier score, a custom ECE calculation with 10 fixed-width probability bins, log loss, and a reliability diagram. The reliability diagram itself uses scikit-learn calibration_curve with 10 quantile bins. PR and ROC curves compare ranking performance. These are distinct diagnostics: strong ranking does not automatically guarantee well-calibrated probabilities.

Model	Holdout Brier	Holdout ECE (10 fixed-width bins)	Holdout log loss	95% bootstrap CI for Brier
Logistic Regression	0.0222	0.0253	0.0786	0.0070–0.0423
Random Forest	0.0290	0.0556	0.1085	0.0144–0.0461
HistGradientBoosting	0.0269	0.0290	0.1059	0.0045–0.0533
Dummy (stratified)	0.2327	0.0052	0.6582	0.2105–0.2571

The Dummy model's low ECE must not be mistaken for useful predictive performance: it predicts a near-constant prior and has ROC-AUC 0.50 and recall 0.00 at threshold 0.50. ECE is bin-dependent and should be interpreted alongside discrimination and proper scoring rules.

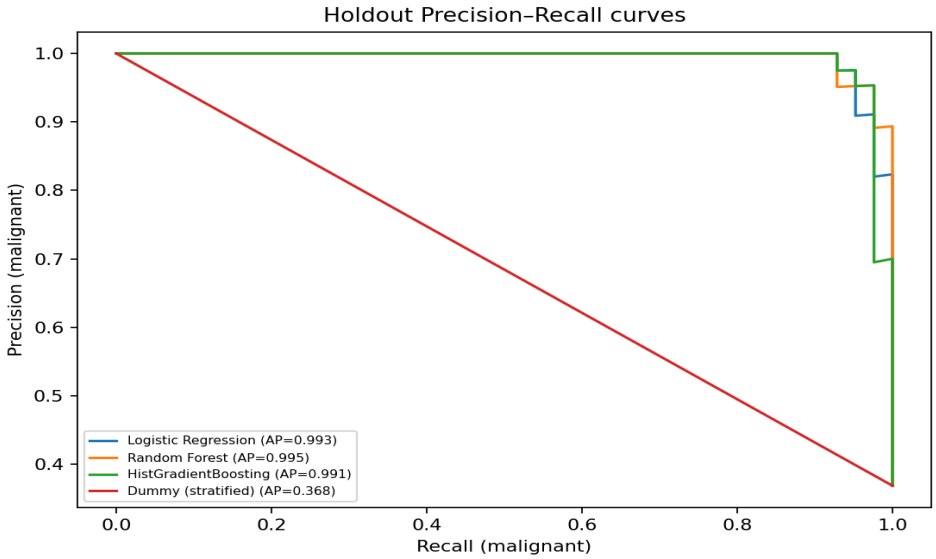


Figure 10. Holdout precision–recall curves.

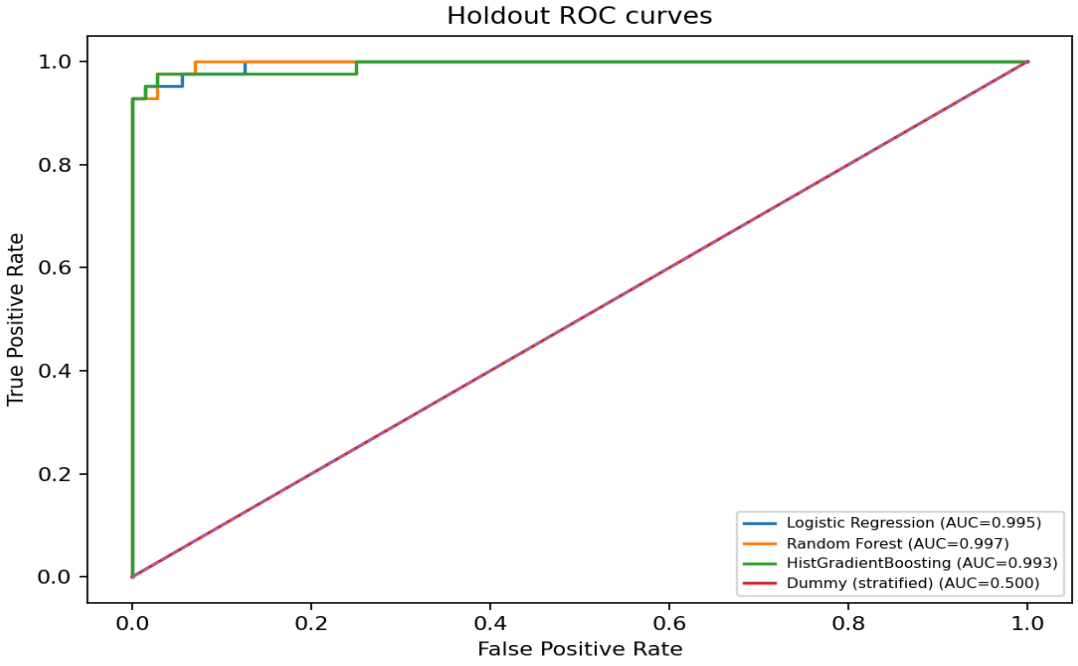


Figure 11. Holdout ROC curves.

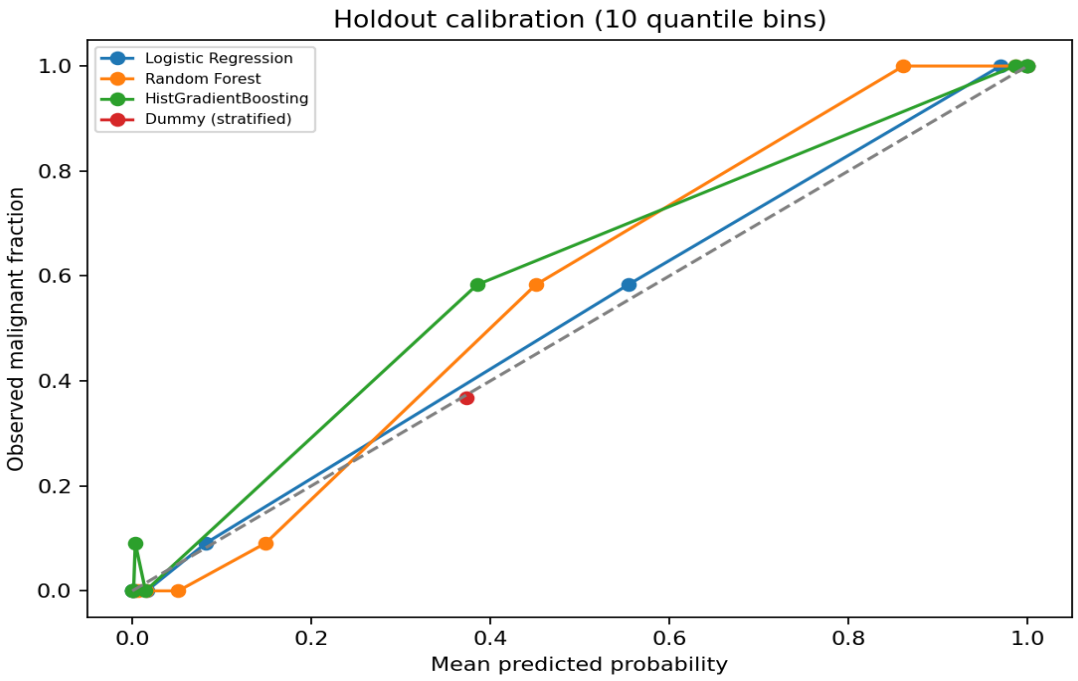


Figure 12. Holdout calibration/reliability diagram (10 quantile bins).

The cost curve is explicitly computed on holdout data as a diagnostic only. The vertical line marks the threshold selected from training OOF probabilities. The holdout cost curve is not used to choose the threshold.

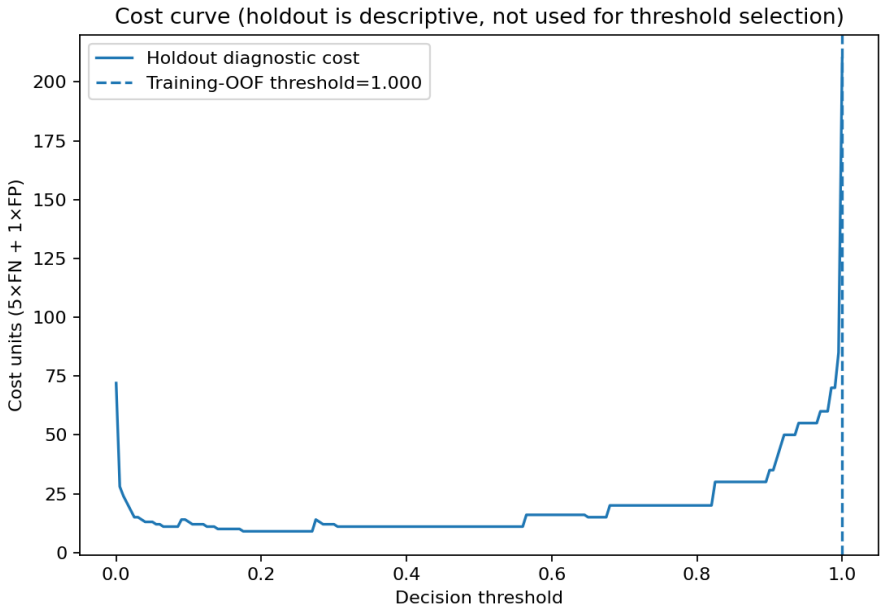


Figure 13. Holdout diagnostic cost curve; not used for threshold selection.

8. Error slices and counterexample

The experiment selects Logistic Regression using training-CV PR-AUC, then uses its holdout probabilities and the threshold selected from training OOF probabilities to define errors. Radius and texture cut points are the corresponding training-set medians. The holdout observations are grouped above versus at/below those training medians.

Slice	Holdout n	Malignant n	Malignant recall	False negatives	False positives
Mean radius: above training median	58	40	0.4750	21	0
Mean radius: at/below training median	56	2	0.0000	2	0
Mean texture: above training median	51	33	0.4545	18	0
Mean texture: at/below training median	63	9	0.4444	5	0

The at/below-median radius slice has only two malignant observations, so its 0.0000 recall is highly unstable and should not be overinterpreted. The two texture slices have similar recall (0.4545 and 0.4444), though their malignant sample sizes differ.

8.1 Most-confident error counterexample

Field	Recorded experiment output
Error type	False Negative (missed malignant)
Predicted malignant probability	0.999532
Mean radius	16.07
Mean texture	19.65
Selected threshold	0.999612
Reason for error	The predicted probability is slightly below the selected threshold, so the case is classified as negative.

The code identifies the most confident error by sorting all misclassified holdout cases in descending order of predicted malignant probability. This counterexample is therefore not an arbitrarily chosen example. It demonstrates that a very high threshold can classify a malignant case as negative even when its predicted probability is extremely close to 1.

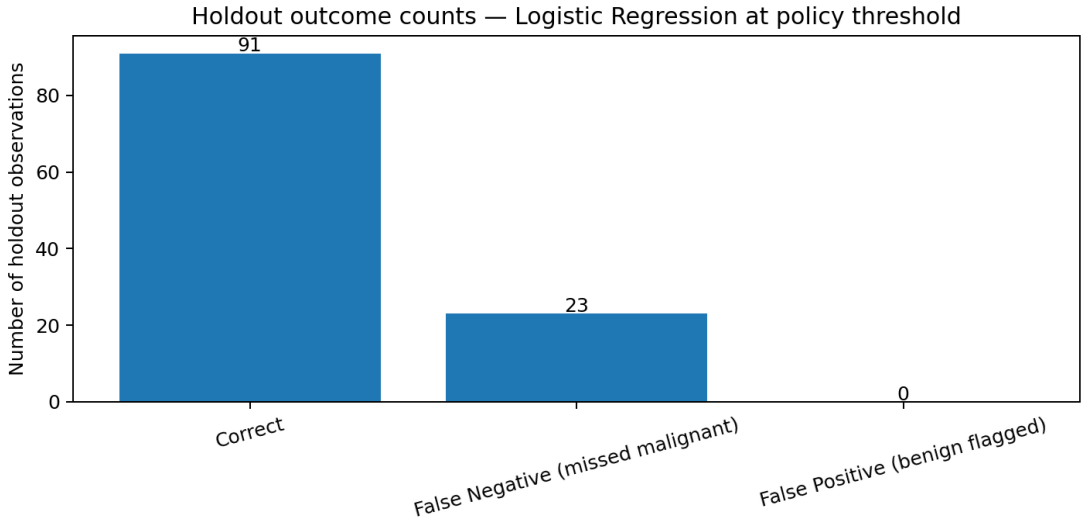


Figure 14. Holdout error-type counts for Logistic Regression at the selected operating threshold.

9. Threshold decision record

The threshold rule minimizes empirical false-negative and false-positive cost subject to a maximum alert rate. The threshold is selected using training out-of-fold probabilities only. The selected model is Logistic Regression because it has the highest mean training-CV PR-AUC.

Decision-record field	Final value
Selected model	Logistic Regression.
Positive class	Malignant (1).
False-negative cost	5.0 cost units.
False-positive cost	1.0 cost unit.
Maximum training alert capacity	20%.
Threshold selection data	Training OOF probabilities only.
Selected threshold	0.999612.
Holdout alert rate	0.166667 (16.7%; 19 of 114).
Holdout confusion counts	TP=19; FP=0; FN=23; TN=72.
Holdout cost	5×23 + 1×0 = 115 cost units.

The experiment also records threshold decisions for every model under the same 5:1 cost ratio and 20% training capacity:

Model	Training-OOF threshold	Holdout alert rate	TP	FP	FN	TN	Holdout cost
Logistic Regression	0.999612	16.7%	19	0	23	72	115
Random Forest	0.977014	19.3%	22	0	20	72	100
HistGradientBoosting	0.998009	19.3%	22	0	20	72	100
Dummy (stratified)	1.000000	0.0%	0	0	42	72	210

For Logistic Regression, the threshold sensitivity table reports the following holdout outcomes as training alert capacity changes, with FP cost fixed at 1:

The holdout alert rate can differ from the training capacity because the threshold is selected on training OOF predictions and then applied to a separate holdout set. The 20% capacity constraint does not imply that exactly 20% of the holdout cases will be alerted.

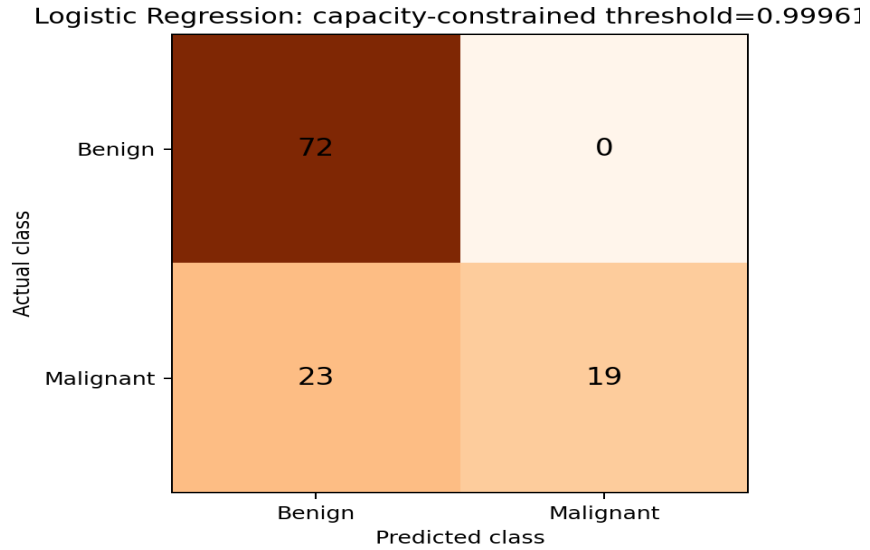


Figure 15. Logistic Regression holdout confusion matrix at the selected threshold.

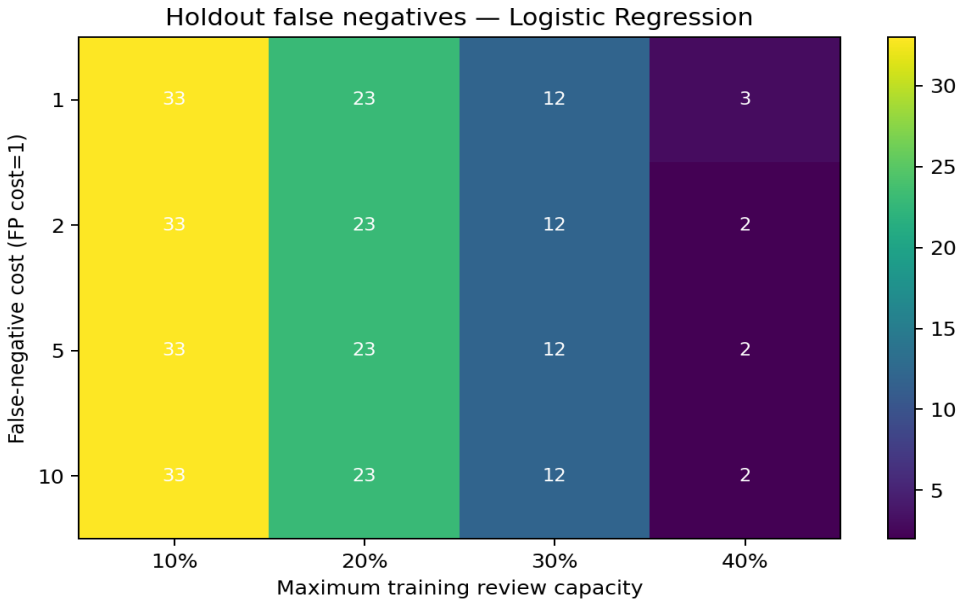


Figure 16. Threshold sensitivity: holdout false negatives across FN costs and training capacities.

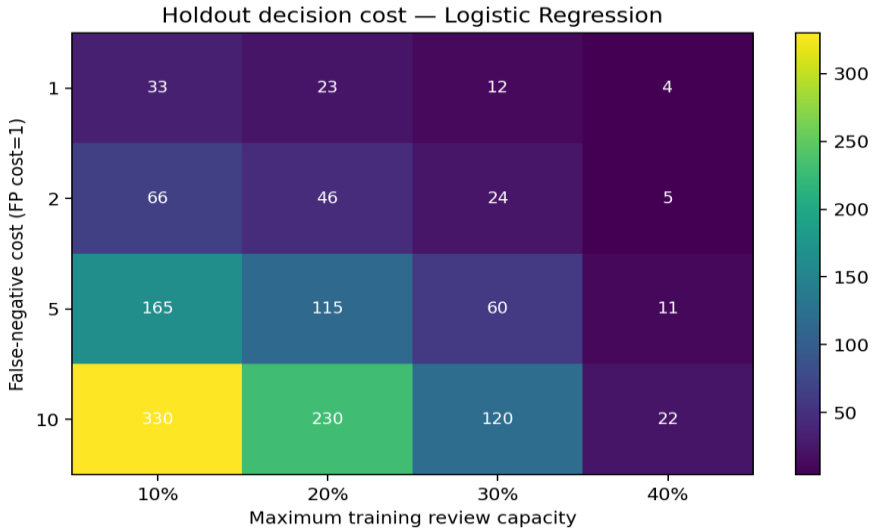


Figure 17. Threshold sensitivity: holdout decision cost.



Figure 18. Threshold sensitivity: holdout alert rate.

Interpretation: the selected policy satisfies the training alert-capacity rule but misses 23 of the 42 malignant holdout observations. This is a substantial sensitivity cost. The cost ratio and capacity are assignment assumptions, not values validated with clinical stakeholders. The policy is therefore an analytical illustration, not a recommended clinical operating threshold.

10. Interpretation and discussion

Logistic Regression wins on mean cross-validation PR-AUC and mean cross-validation Brier score. Random Forest's paired OOF PR-AUC difference is negative with a 95% interval below zero, and its Brier difference is positive with an interval above zero, indicating worse OOF PR-AUC and Brier under this comparison. HistGradientBoosting paired intervals include zero for ROC-AUC, PR-AUC, and Brier, so the experiment does not establish a clear paired difference for those metrics.

Random Forest has slightly higher holdout ROC-AUC and PR-AUC point estimates than Logistic Regression. However, the protocol requires selection using training-CV PR-AUC only. Choosing Random Forest after inspecting holdout metrics would violate the frozen evaluation protocol and introduce holdout-based model selection. Whereas Logistic Regression has the best Brier score and log loss among the learned models on holdout. Yet the capacity-constrained threshold is extremely high. The result illustrates that a model can rank cases well and still have a poor operating point when the review capacity is restrictive. The error slices show that false negatives are concentrated in the above-median radius and texture groups in absolute count, but slice sizes and malignant prevalence differ. The counterexample further demonstrates that thresholding converts a near-one probability into a negative class decision because it falls just below the selected threshold. For this benchmark and frozen protocol, Logistic Regression is the selected model. The experiment does not justify claims of clinical suitability or real-world risk estimation. The primary contribution is a transparent comparison of traditional classifiers, paired uncertainty, calibration diagnostics, cost-sensitive threshold selection, and explicit error analysis.